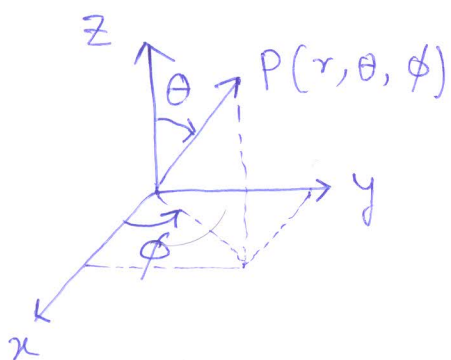




Legendre Polynomials

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + \alpha(\alpha+1)y = 0$$

$$\Rightarrow \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \alpha(\alpha+1)y = 0$$

α is real and non-negative

$$y(x) = C_1 P_n(x) + C_2 Q_n(x) \quad \text{where } n = \alpha$$

$\uparrow$ Legendre polynomials of degree n $\uparrow$ Legendre functions of 2nd kind
 $= \text{zero or positive integer}$

$$P_0(x) = 1$$

$$P_n(x) = (-1)^{n/2} \frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n} U_n(x), \quad n=2,4,6,\dots$$

and $P_1(x) = x$

$$P_n(x) = (-1)^{(n-1)/2} \frac{1 \cdot 3 \cdot 5 \cdots n}{2 \cdot 4 \cdot 6 \cdots (n-1)} V_n(x), \quad n=3,5,7,\dots$$

Here

$$U_n(x) = 1 - \frac{n(n+1)}{2} x^2 + \frac{n(n-2)(n+1)(n+3)}{24} x^4 - \dots$$

$$V_n(x) = x - \frac{(n-1)(n-2)}{6} x^3 + \frac{(n-1)(n-3)(n+2)(n+4)}{120} x^5 - \dots$$

If $n = \text{even positive integer}$, $U_n \rightarrow x^n$

If $n = \text{odd}$, $V_n \rightarrow x^n$

Otherwise, these expressions are infinite series convergent only when $|x| < 1$

First few Legendre Polynomials

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x), \quad P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

$$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$$

The Legendre Polynomials can also be expressed by Rodrigue's formula:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Useful recurrence formula:

$$P_0(x) = \frac{1}{x} P_1(x), \quad P_1(x) = \frac{1}{3x} [P_0(x) + 2P_2(x)]$$

In general:

$$P_n(x) = \frac{n P_{n-1}(x) + (n+1) P_{n+1}(x)}{(2n+1)x}$$

If $|x| < 1$, the Legendre function of the 2nd kind are given by:

$$Q_0(x) = V_0(x)$$

$$Q_n(x) = (-1)^{n/2} \frac{2 \cdot 4 \cdot 6 \cdots n}{1 \cdot 3 \cdot 5 \cdots (n-1)} V_n(x), \quad n=2,4,6,\dots$$

and

$$Q_1(x) = -U_1(x)$$

$$Q_n(x) = (-1)^{(n+1)/2} \frac{2 \cdot 4 \cdot 6 \cdots (n-1)}{1 \cdot 3 \cdot 5 \cdots n} U_n(x), \quad n=3,5,7,\dots$$

Legendre function of 2nd kind are infinite series, which are convergent when $|x| < 1$, but diverges as $x \rightarrow \pm 1$

Fourier-Legendre Series

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \lambda(\lambda+1)y = 0$$

$$P(x) = 1-x^2, \quad q(x) = 0, \quad w(x) = 1, \quad \lambda = \lambda(\lambda+1)$$

NOTE: The function $P(x)$ vanishes at $x = \pm 1$

$\Downarrow$

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0, \quad m \neq n$$

$$f(x) = \sum_{n=0}^{\infty} A_n P_n(x), \quad -1 < x < 1$$

$$A_n = \frac{\int_{-1}^1 f(x) P_n(x) dx}{\int_{-1}^1 [P_n(x)]^2 dx}$$

$\swarrow \quad \frac{2}{2n+1}$

Alternatively,

$$A_n = \frac{2n+1}{2^{n+1} \lfloor n} \int_{-1}^{+1} (1-x^2)^n \frac{d^n f(x)}{dx^n} dx$$

$P_n(x)$ is an even function of x when n is even
 $P_n(x)$ is an odd function of x when n is odd

For even function $f(x)$

$$A_n = \begin{cases} 0 & n = \text{odd} \\ (2n+1) \int_0^1 f(x) P_n(x) dx & n = \text{even} \end{cases}$$

For odd function $f(x)$

$$A_n = \begin{cases} (2n+1) \int_0^1 f(x) P_n(x) dx & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$$

Expansion valid in $(-a, +a)$

Replace x by $\frac{x}{a}$

$$f(x) = \sum_{n=0}^{\infty} A_n P_n\left(\frac{x}{a}\right), \quad -a < x < a$$

Where
$$A_n = \frac{2n+1}{2a} \int_{-a}^{+a} f(x) P_n\left(\frac{x}{a}\right) dx$$

or
$$A_n = \frac{2n+1}{2^{n+1} \ln a^{n+1}} \int_{-a}^{+a} (a^n - x^n)^n \frac{d^n f(x)}{dx^n} dx$$